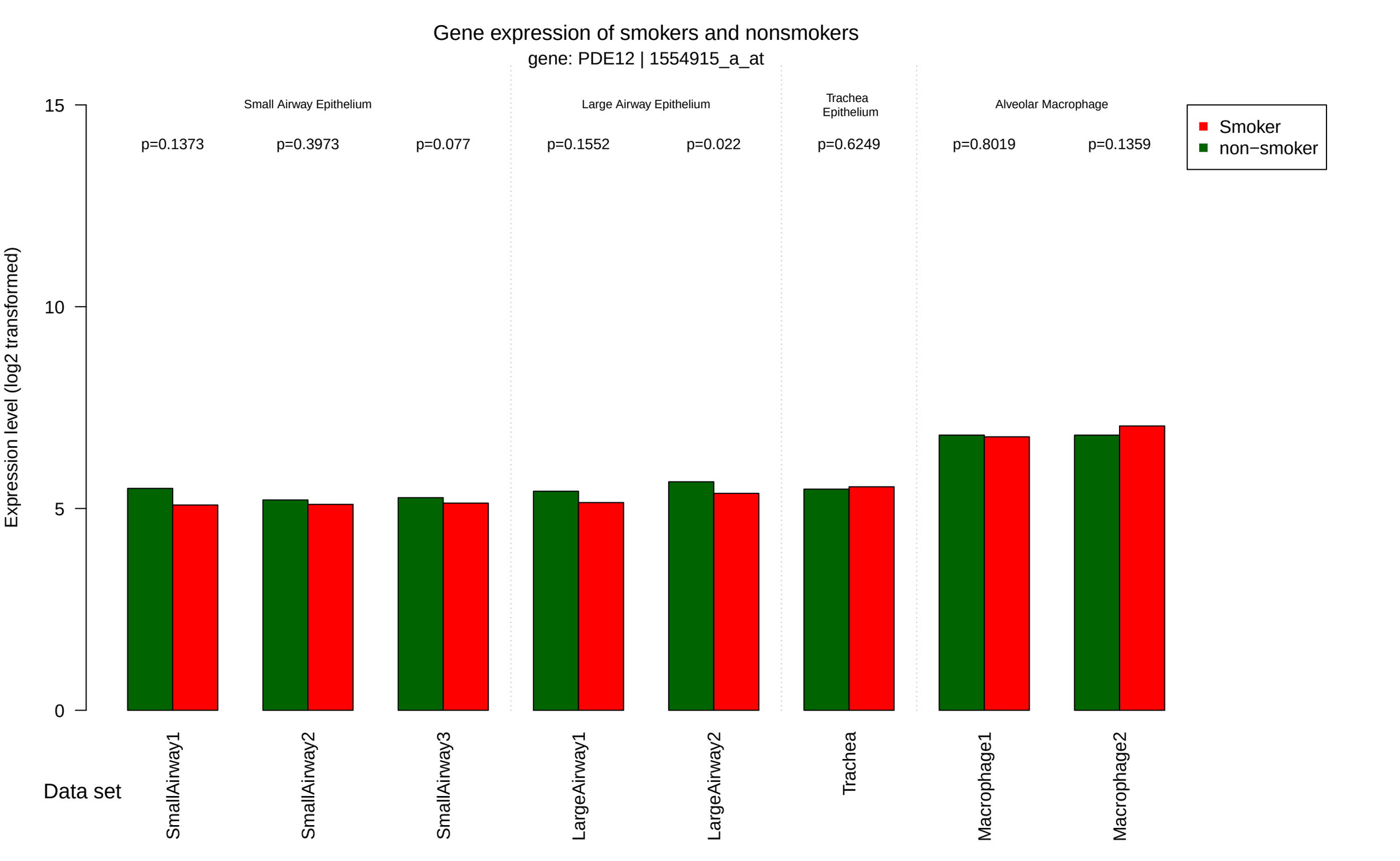
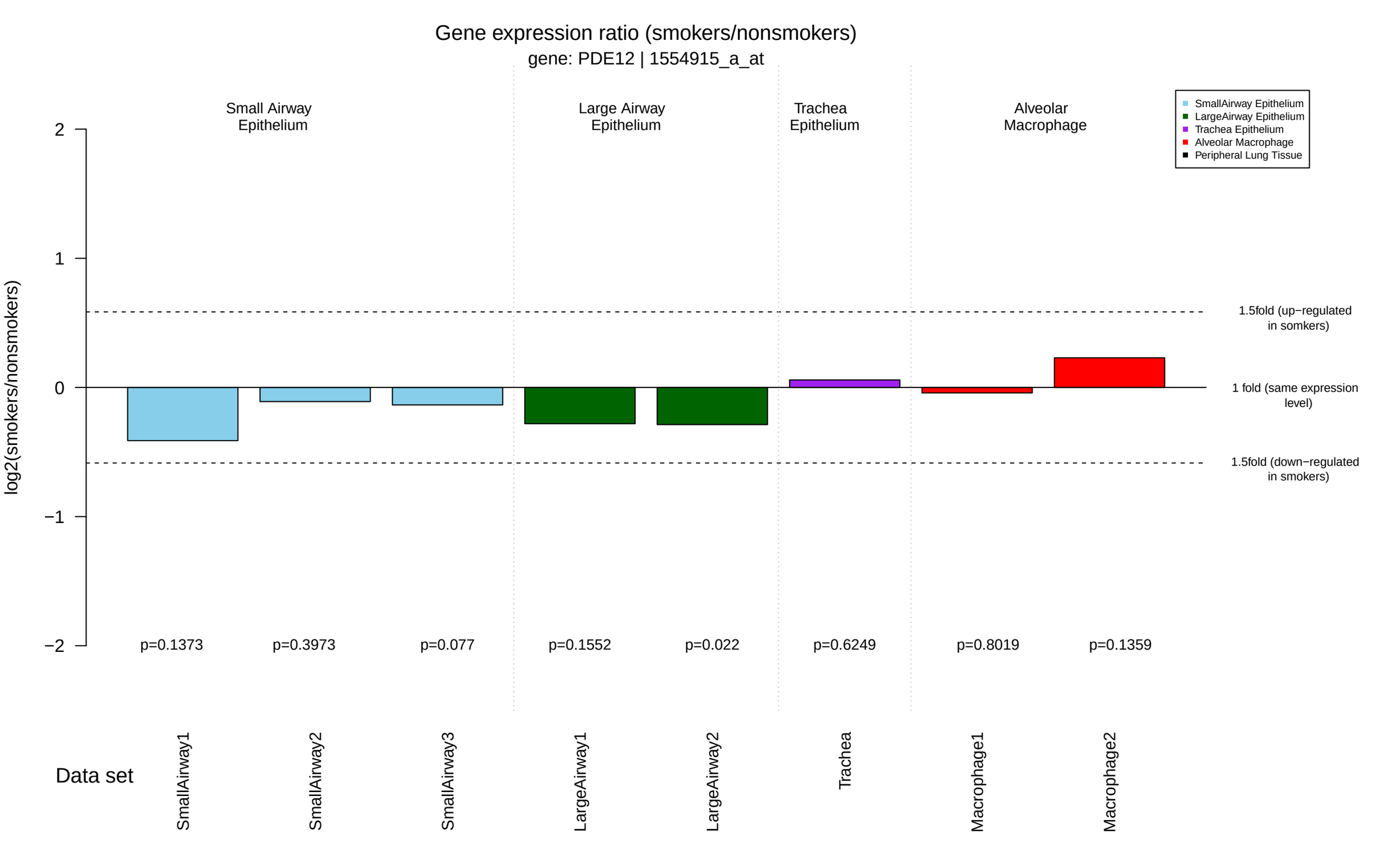
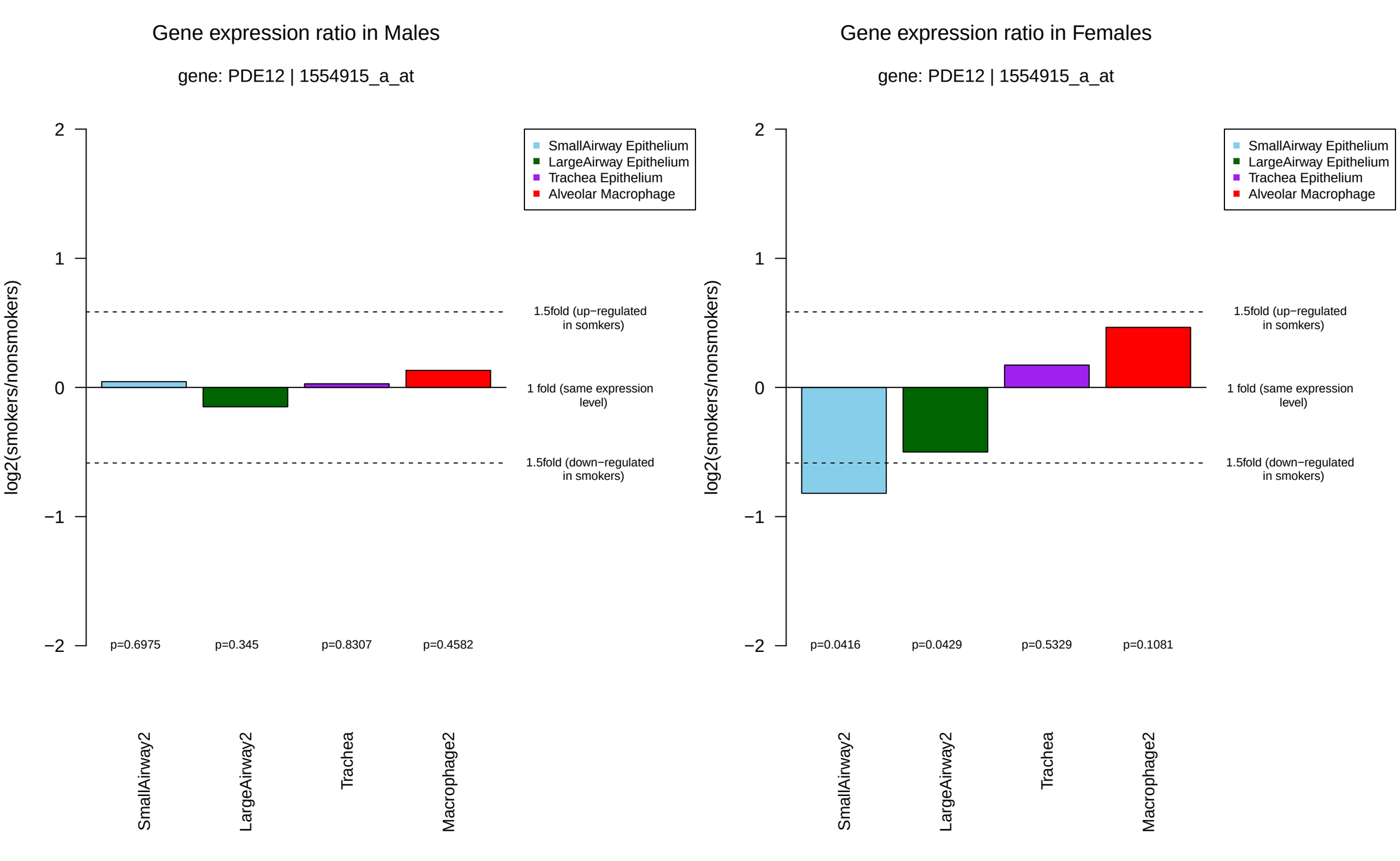
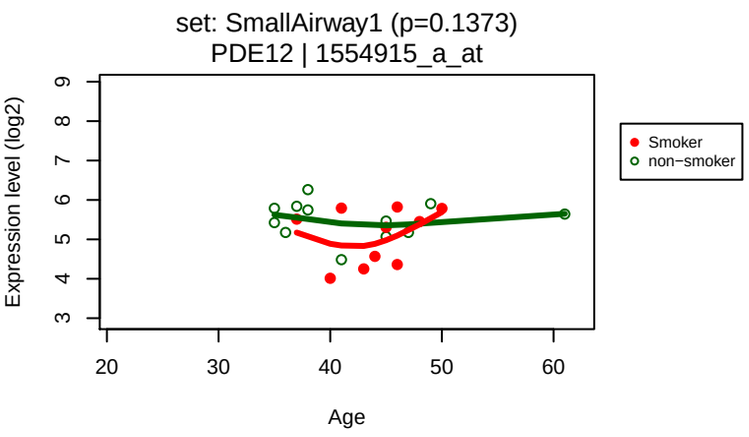
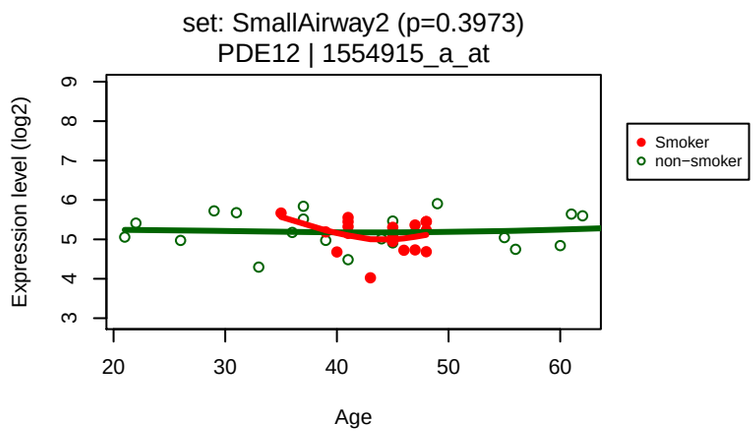
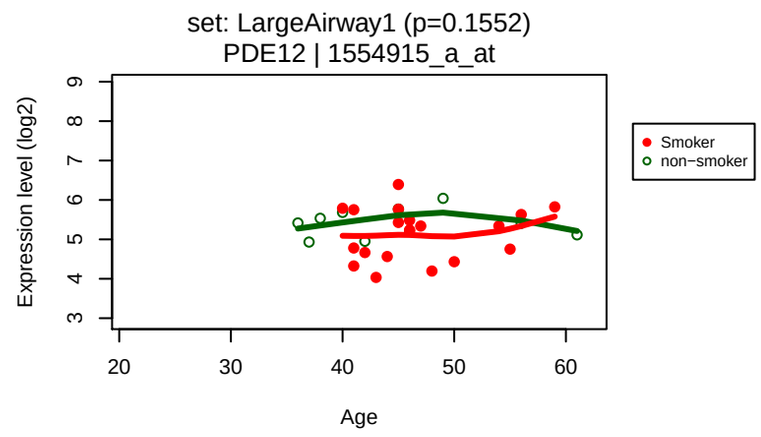

Gene expression of smokers and nonsmokers
gene: PDE12 | 1554915_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1373
p=0.3973
p=0.077
p=0.1552
p=0.022
p=0.6249
p=0.8019
p=0.1359
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

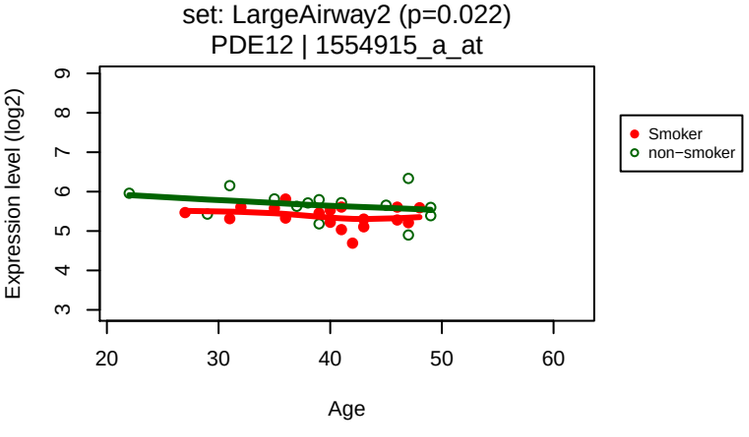

Gene expression ratio (smokers/nonsmokers)
gene: PDE12 | 1554915_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1373
p=0.3973
p=0.077
p=0.1552
p=0.022
p=0.6249
p=0.8019
p=0.1359
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

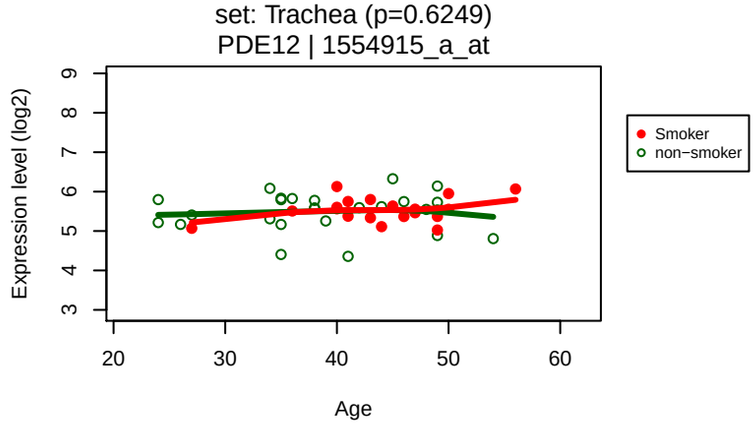

Gene expression ratio in Males
gene: PDE12 | 1554915_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6975
p=0.345
p=0.8307
p=0.4582
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE12 | 1554915_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0416
p=0.0429
p=0.5329
p=0.1081
SmallAirway2
LargeAirway2
Trachea
Macrophage2

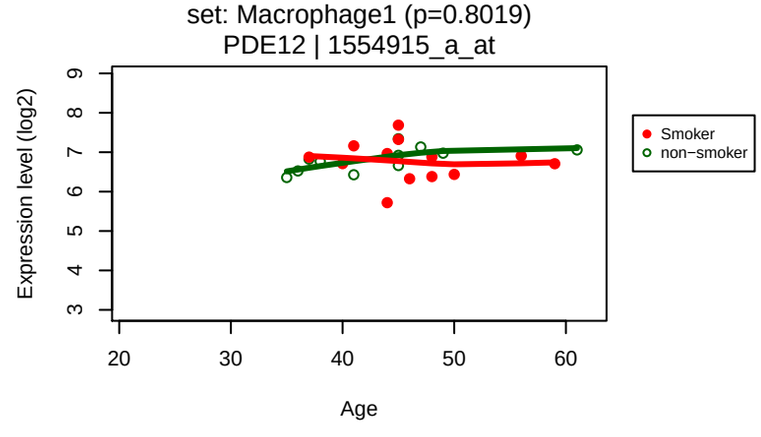

set: SmallAirway1 (p=0.1373)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker

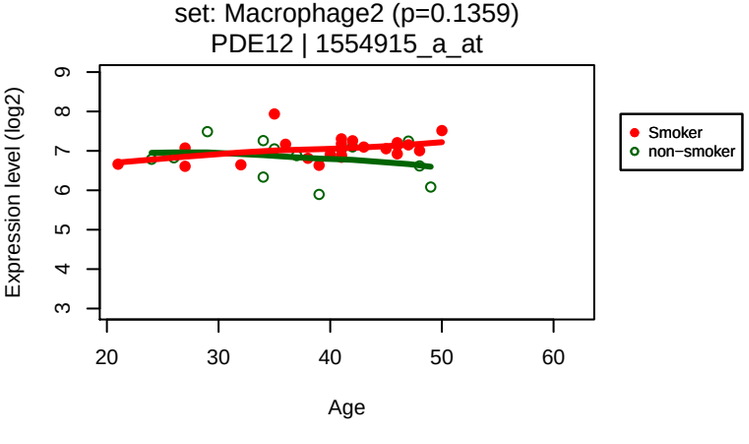

set: SmallAirway2 (p=0.3973)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker

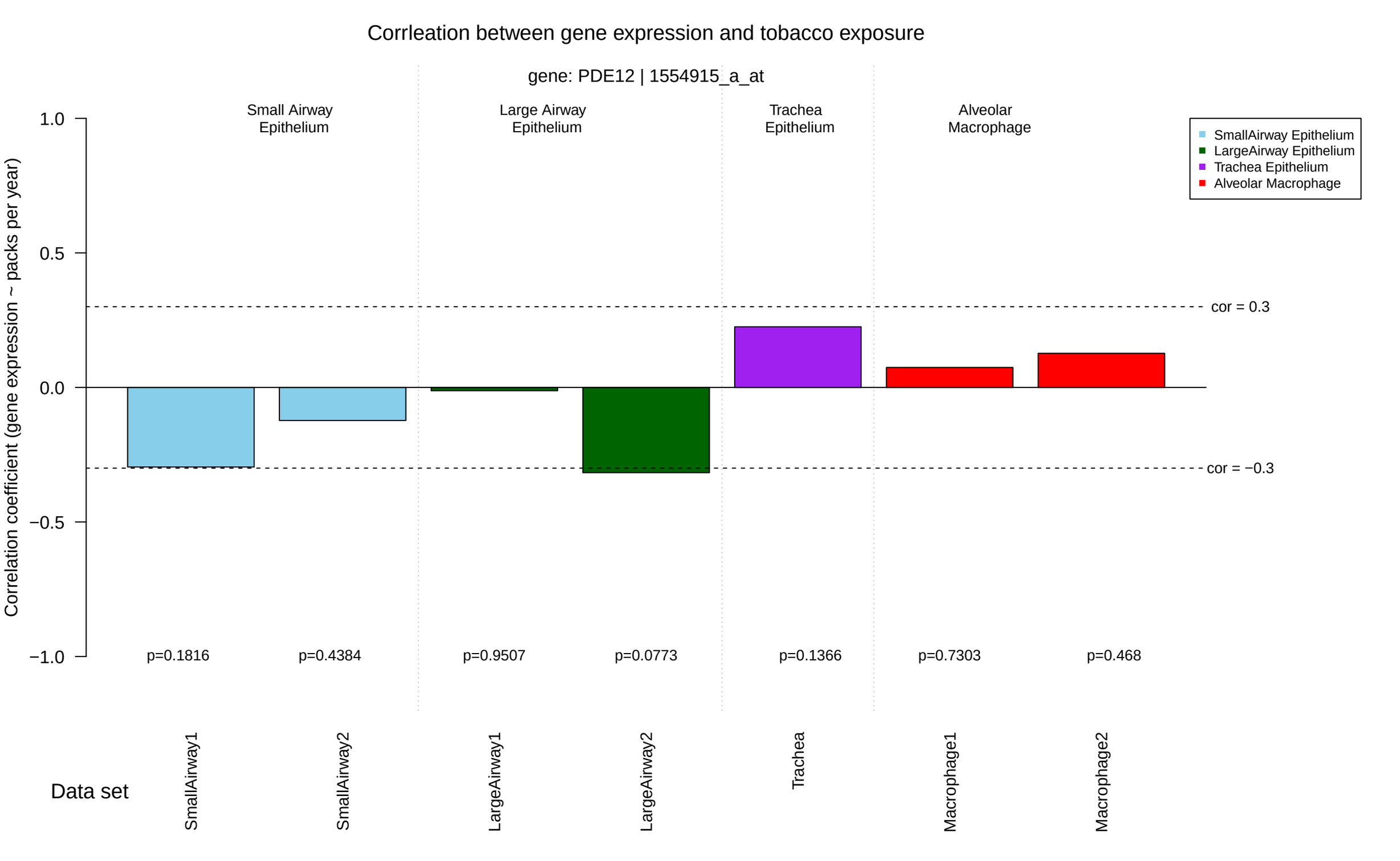

set: LargeAirway1 (p=0.1552)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.022)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6249)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8019)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1359)
PDE12 | 1554915_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE12 | 1554915_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1816
p=0.4384
p=0.9507
p=0.0773
p=0.1366
p=0.7303
p=0.468
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2